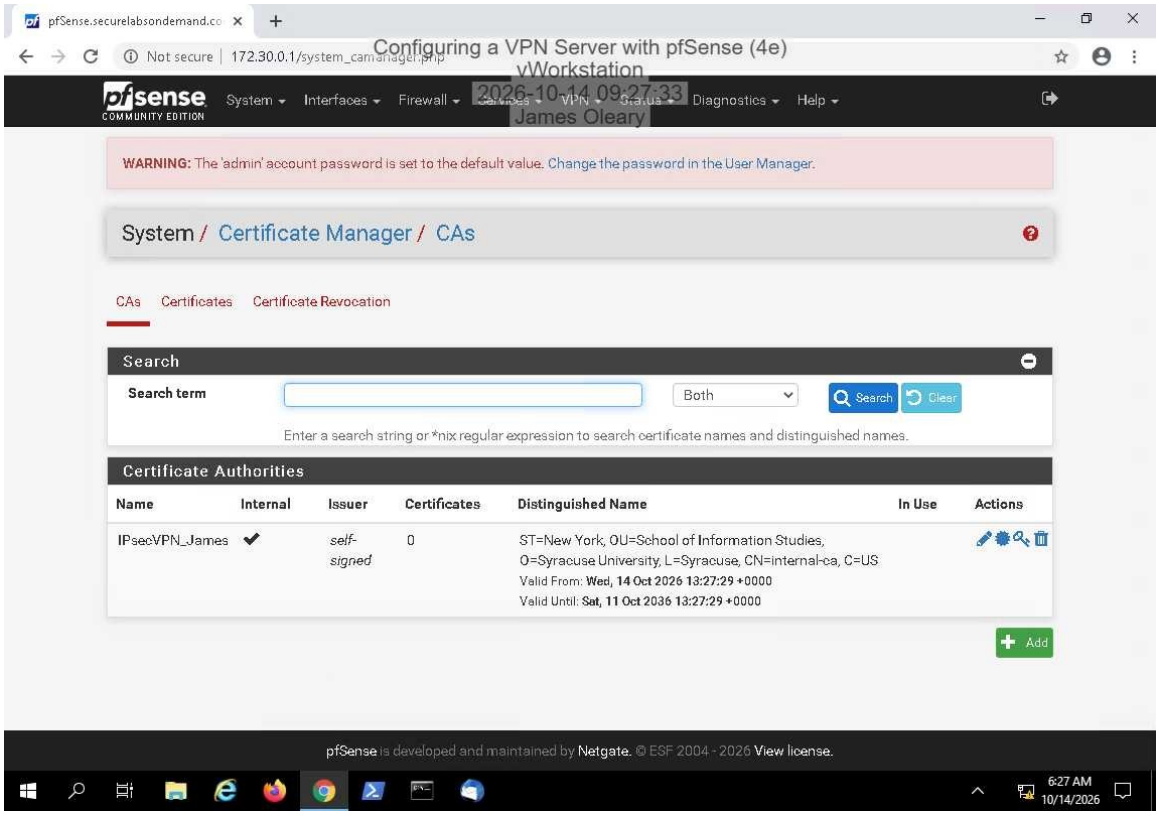
Lab 2: Configuring a VPN Server with pfSense

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Hands-On Demonstration Part 1: Configure an IPsec VPN Server

Figure 1

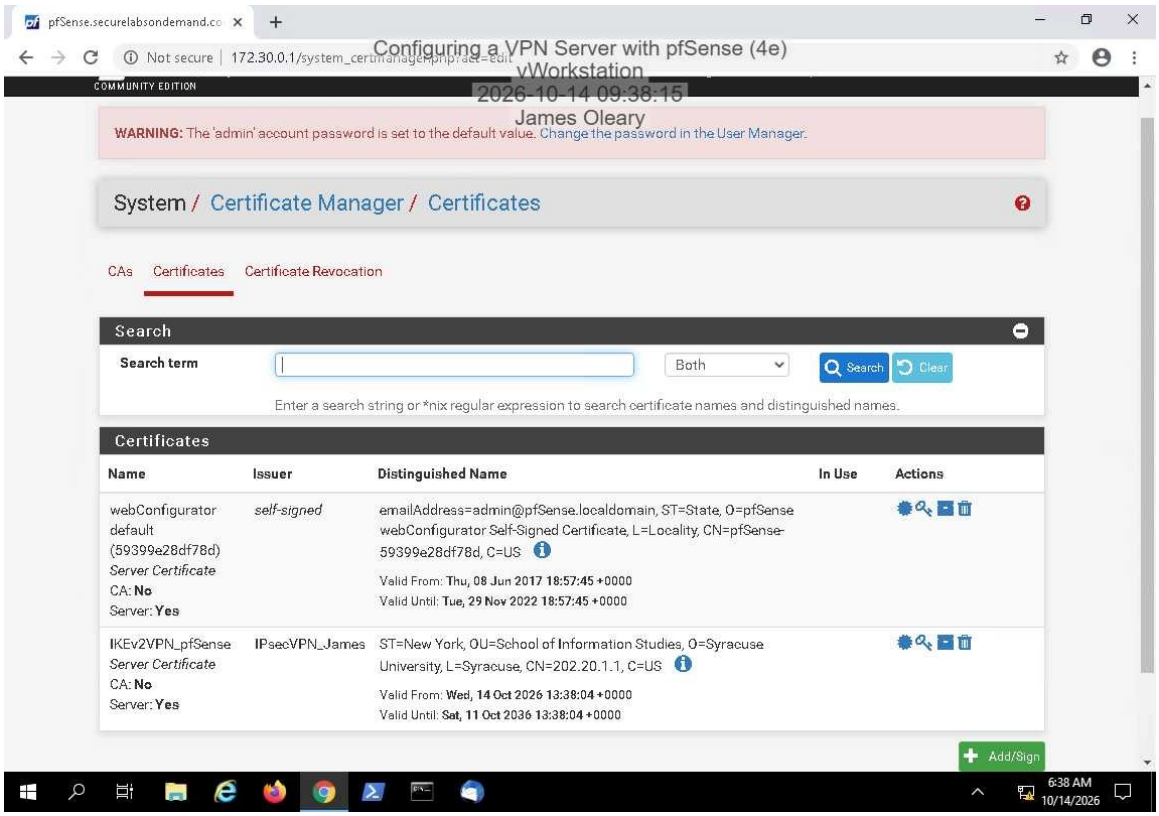
Required Screenshot 1: Updated Certificate Authorities Table



Note. The Certificate Authorities table shows the new CA "IPsecVPN_James", which signs the VPN server certificate.

Figure 2

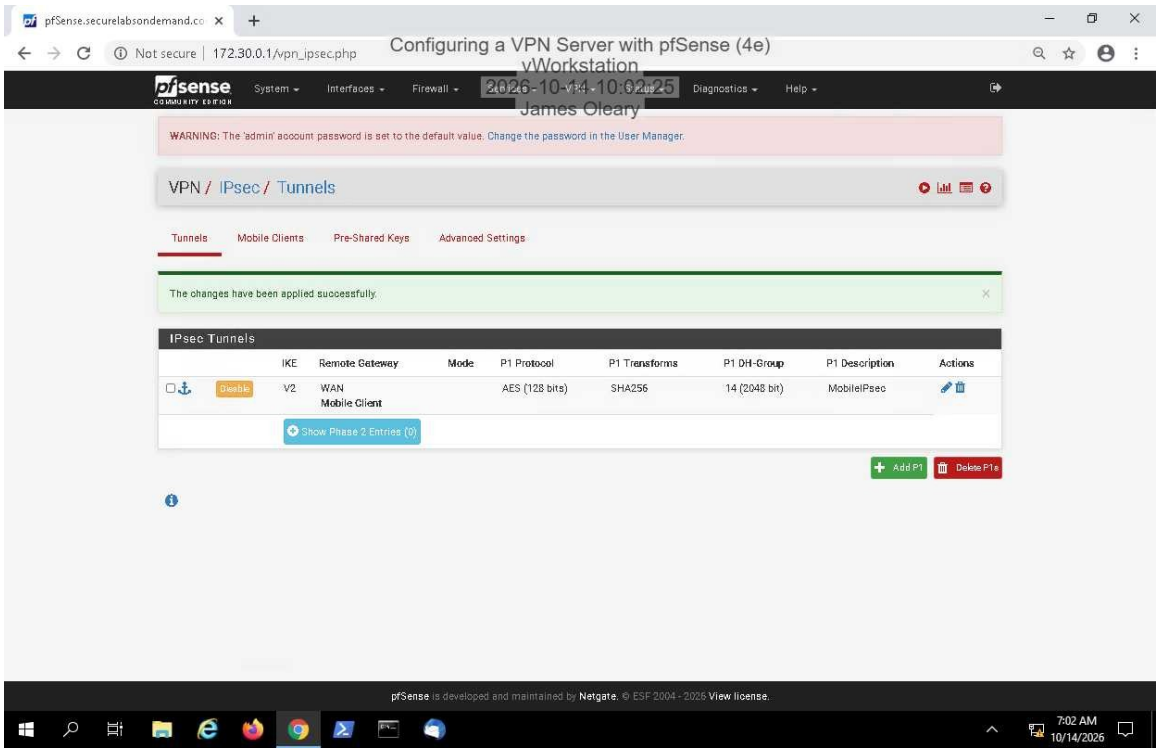
Required Screenshot 2: Updated Certificates Table



Note. The Certificates table shows the server certificate "IKEv2VPN_pfSense" issued by the new CA "IPsecVPN_James".

Figure 3

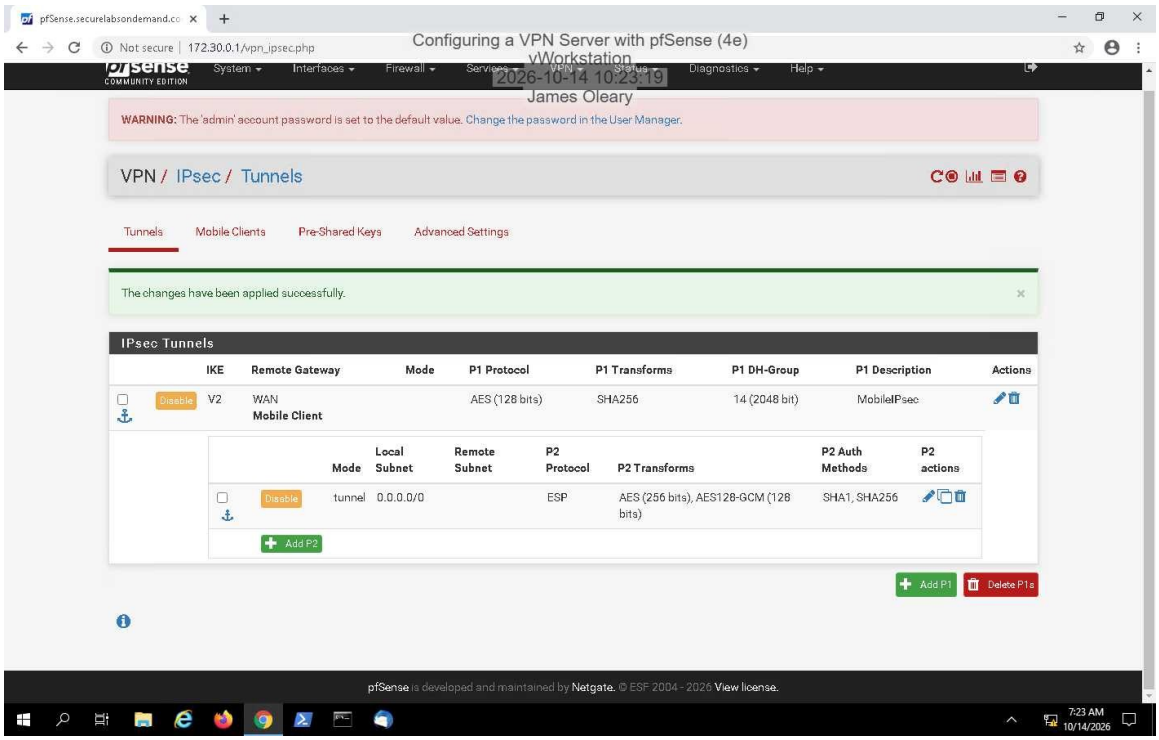
Required Screenshot 3: IPsec Tunnels Table After Phase 1



Note. The IPsec Tunnels table after the Phase 1 (IKE) mobile-client configuration was saved.

Figure 4

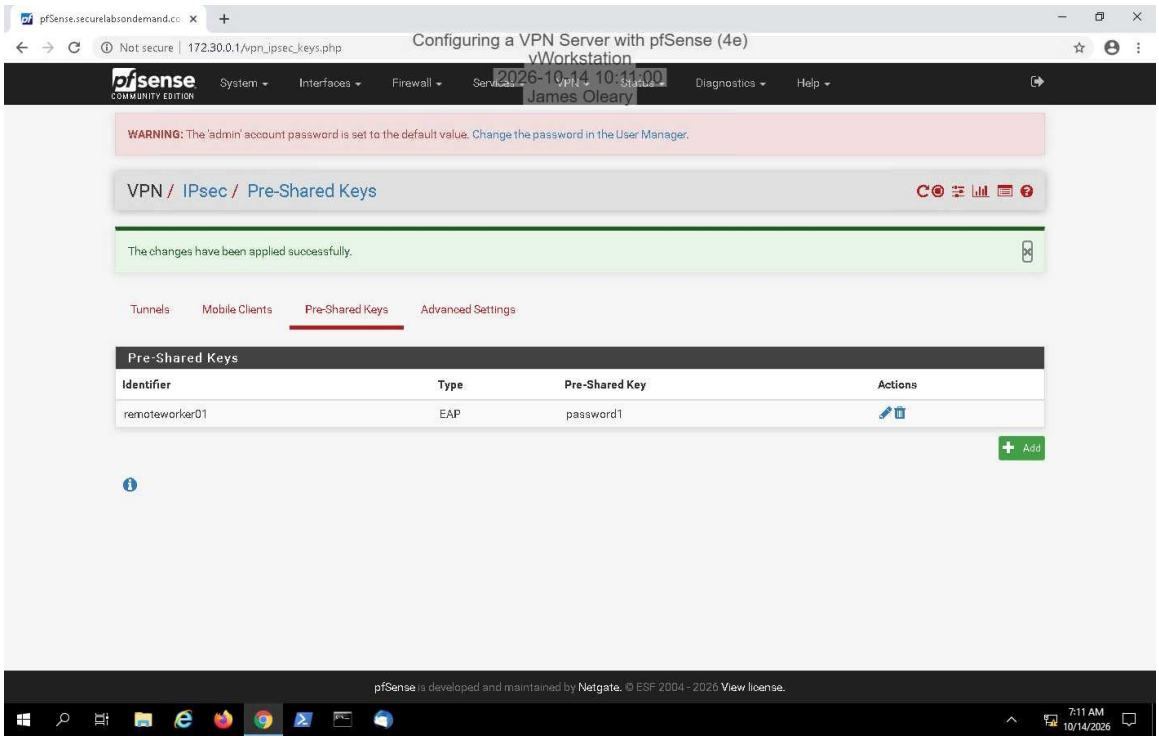
Required Screenshot 4: IPsec Tunnels Table After Phase 2



Note. The IPsec Tunnels table after the Phase 2 (IPsec) entry was added, in Tunnel mode. This screenshot was recaptured after Screenshot 5, which is why its timestamp is later.

Figure 5

Required Screenshot 5: Updated Pre-Shared Keys Table

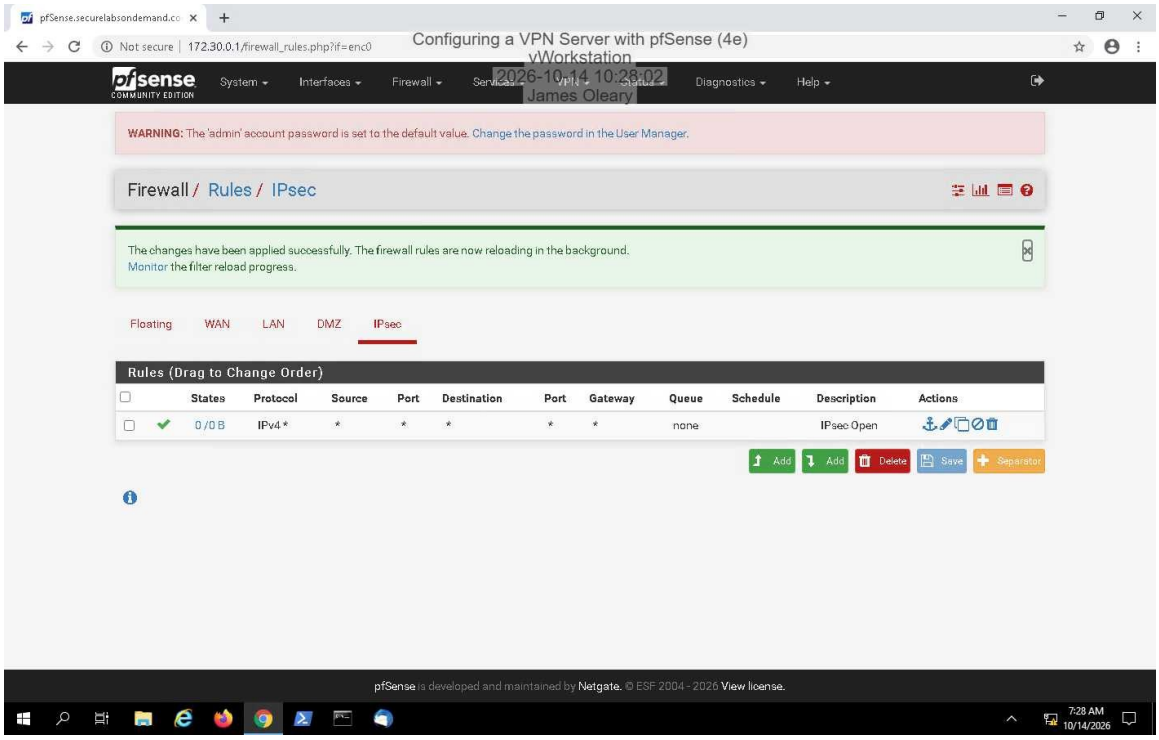


Note. The Pre-Shared Keys table shows the assigned authentication entry for user remoteworker01.

Hands-On Demonstration Part 2: Configure a Firewall Rule for VPN Traffic

Figure 6

Required Screenshot 6: Updated IPsec Rules Table



Note. The Firewall Rules table on the IPsec interface shows the new rule that permits all IPv4 traffic from remote VPN users.

Research Question 1: IPsec Tunnel Mode and Transport Mode

> *IPsec supports two operating modes, Tunnel Mode and Transport Mode. How does each mode function, and what are the trade-offs between them?*

The two IPsec modes differ in how much of the original packet they protect. In tunnel mode, ESP encrypts the entire original IP packet, header and payload, then wraps it inside a new IP packet whose header carries the addresses of the two VPN endpoints (Kent & Seo, 2005). This makes it the standard choice for site-to-site VPNs linking whole subnets across the internet. Our lab used tunnel mode for Phase 2 because remote workers' traffic also crosses the untrusted internet, and hiding the inner addresses keeps an eavesdropper from seeing which internal host they reach (Barker et al., 2020). Transport mode encrypts only the payload and leaves the

original IP header in place, so it fits host-to-host communication, such as two servers on an internal network (Kent, 2005).

The trade-offs follow from that design. Tunnel mode protects the path between gateways and makes traffic analysis harder, since an observer sees only the gateway addresses. Transport mode protects the path between hosts but exposes source, destination, and timing, which can reveal what service a user is trying to reach. Transport mode is also lighter, since it adds no outer header. Tunnel mode handles NAT better, because a NAT device rewrites only the outer header and NAT traversal can wrap ESP in UDP port 4500, while transport mode can break when NAT alters the original addresses (Barker et al., 2020). Tunnel mode is the better fit for traffic crossing untrusted networks, and transport mode for host-to-host links inside a trusted network.

References

Barker, E., Dang, Q., Frankel, S., Scarfone, K., & Wouters, P. (2020). Guide to IPsec VPNs (NIST Special Publication 800-77 Rev. 1). National Institute of Standards and Technology. https://doi.org/10.6028/NIST.SP.800-77r1

Kent, S. (2005). IP Encapsulating Security Payload (ESP) (RFC 4303). Internet Engineering Task Force. https://doi.org/10.17487/RFC4303

Kent, S., & Seo, K. (2005). Security architecture for the Internet Protocol (RFC 4301). Internet Engineering Task Force. https://doi.org/10.17487/RFC4301